Machine Learning

M1 : Introduction

BITS Pilani
Pilani Campus

Dr. Pankaj Agarwal, Professor & Guest faculty

Profile : Dr. Pankaj Agarwal

Total Experience: More than 23 years of Experience in Academics & Industry.
Current Status: Guest Faculty at BITS Pilani **and** Professor & Dean (Engineering)
Areas of interest & research: Data Science & Machine Learning algorithms, Algorithms Design & Analysis, Data Analytics, Soft computing Techniques, NLP, Computer Vision & Deep Learning with Python Programming, Tableau, KNIME etc
Research Publications: more than 60 research publications in international journals/conferences.
Patents: 10 research patents including Australian, German & Indian patents.
Certifications: IBM certified data Science Professional & Microsoft certified Programmer.
Consultant & Trainer: BITS Pilani, Ureka International, Princeton Hive, Great Learning.
Books: Design & Analysis of Algorithms, .NET,SPM, RDBMS, C programming, Computer Organization, Cyber Security
Research Guidance: more than 100 M.Tech Students (BITS Pilani), 5 Ph.D students completed

Disclaimer and Acknowledgement

- The content for these slides has been obtained from books and various other source on the Internet
- I hereby acknowledge all the contributors for their material and inputs.
- I have provided source information wherever necessary
- I have added and modified the content flow to suit the requirements of the course and for ease of class presentation
- Students are requested to refer to the textbook and detailed content of this presentation deck over canvas

Course Introduction

- **Objective of course**
  - Introduction to the basic concepts and techniques of Machine Learning
  - Gain experience in basics of doing independent study and research in the field of Machine Learning
  - Develop skills of using recent machine learning software tools to evaluate learning algorithms and model selection for solving practical problems
- **Focus of this course**
  - Strong Mathematical Foundations of ML algorithms
  - Structured Data Analytics
  - IDD (Independent & Identically Distributed Data)
- **Topics not expected of this course**
  - Unstructured Data Analytics
  - Time Series/Sequence Data Analytics
  - Deep Learning

Course Plan

M1	Introduction
M2	Machine learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification
M5	Decision Tree
M6	Instance Based Learning
M7	Support Vector Machine
M8	Bayesian Learning
M9	Ensemble Learning
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

Text books and Reference book(s)

T1 Tom M. Mitchell: Machine Learning, The McGraw-Hill Companies

R1 Christopher M. Bishop: Pattern Recognition & Machine Learning, Springer

P. Tan, et al. Introduction to Data Mining, Pearson

R2 C.J.C. BURGES: A Tutorial on Support Vector Machines for Pattern

R3 Recognition, Kluwer Academic Publishers, Boston.

Evaluation scheme

- **Quiz (10% - Best 2 of 3 quizzes)**
- **Assignment (20% - 1 Progressive Group Assignment)**
- **Mid-semester exam (30%)**
- **Comprehensive exam (40%)**

Pre-requisites

- Linear algebra: vector/matrix manipulations, properties
- Calculus: partial derivatives
- Probability: common distributions; Bayes Rule
- Statistics: mean/median/mode; maximum likelihood

Lab Plan

Lab No.	Lab Objective
1	End to End Machine Learning
2	Linear Regression and Gradient Descent Algorithm
3	Logistic Regression Classifier
4	Decision Tree
5	Naïve Bayes Classifier
6	Random Forest

- **Labs not graded**
- **Most of the Lab recordings available at CSIS virtual labs**
- **Webinars will be conducted for lab sessions**
- **Labs will be conducted in Python**

Agenda

- What is Machine Learning?
- Why Machine Learning is important?
- Types of Machine Learning
- Application Areas
- Issues in Machine Learning
- Demo Case study

What is Machine Learning?

- The term Machine Learning was first coined by **Arthur Samuel** in the year 1959, an IBM employee and pioneer in the field of computer gaming and artificial intelligence
- Humans can learn from their experiences with their learning capability,
- But can a machine also learn from experiences or past data like a human does?
- It is the branch of AI mainly concerned with the development of algorithms which allow a computer to learn from the **data** and **past experiences** on their own.
- Machine Learning is a system that can learn from **example** through **self-improvement** and without being explicitly coded by programmer.
- Machine learning uses various algorithms for building **mathematical models** and making **predictions** using historical data or information

What is Machine Learning?

Definition by Tom Mitchell (1998):

"A computer program is said to learn from **experience E** with respect to some class of **tasks T** and **performance measure P**, if its performance at tasks in T, as measured by P, **improves with experience E**."

Example: playing checkers.

E = the experience of playing many games of checkers

T = the task of playing checkers.

P = the probability that the program will win the next game.

What is Machine Learning?

- To have a learning problem, we must identify
  - The class of tasks
  - The measure of performance to be improved
  - Source of experience

Modern-day machine learning has two objectives, one is to **classify data** based on models which have been developed, the other purpose is to **make predictions** for future outcomes based on these models.

Introduction to Machine Learning

Defining the Learning Tasks

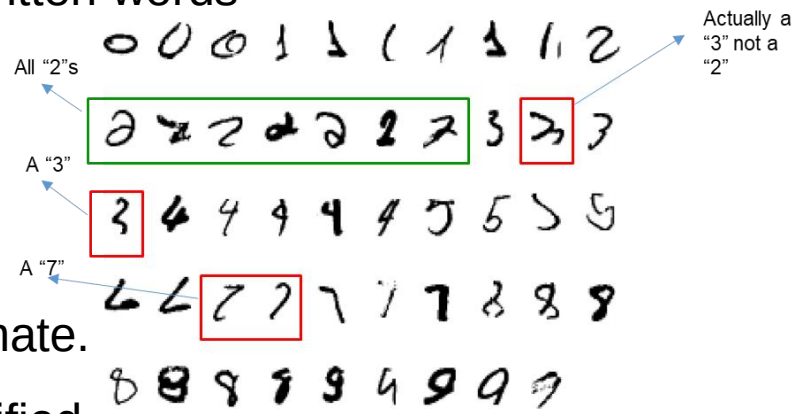
Improve on task T, with respect to performance metric P, based on experience E

Example 1

T: Recognizing hand-written words

P: Percentage of words correctly classified

E: Database of human labelled images of handwritten words



Example 2

T: Categorize email messages as spam or legitimate.

P: Percentage of email messages correctly classified.

E: Database of emails, some with human-given labels

Introduction to Machine Learning

Defining the Learning Tasks

Improve on task T, with respect to performance metric P, based on experience E

Example 3

T: Playing Checkers

P: Percent of games won against opponents

E: Games Played against itself

Example 4

T: Drive on public four-lane highways using vision sensors.

P: Average distance travelled before an error (as judged by human).

E: A sequence of images and steering commands recorded while observing a human driver

Introduction to Machine Learning

Applications - Perspectives

Object Categorization

- Medical Diagnosis
- Transaction Analysis
- Recommendation System
- Speech – Text Processing

Sequence Prediction

- Forecasting
- Medical Research
- Recommendation System
- Content Management

Introduction to Machine Learning

Applications - Perspectives

Planning

Navigation

Path Finding

Gaming

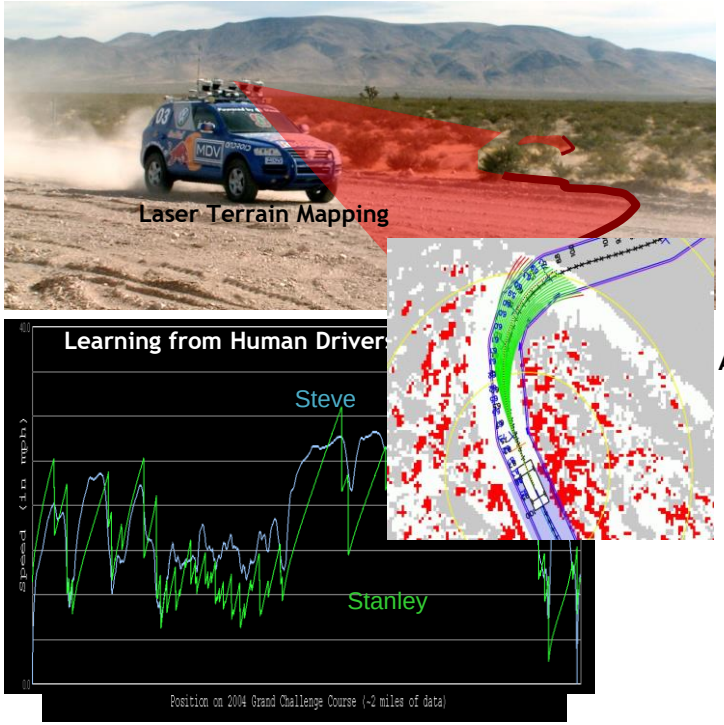
Controlling

Problem Solving

Pattern Recognition

Optimization

Decision System



Adaptive Vision

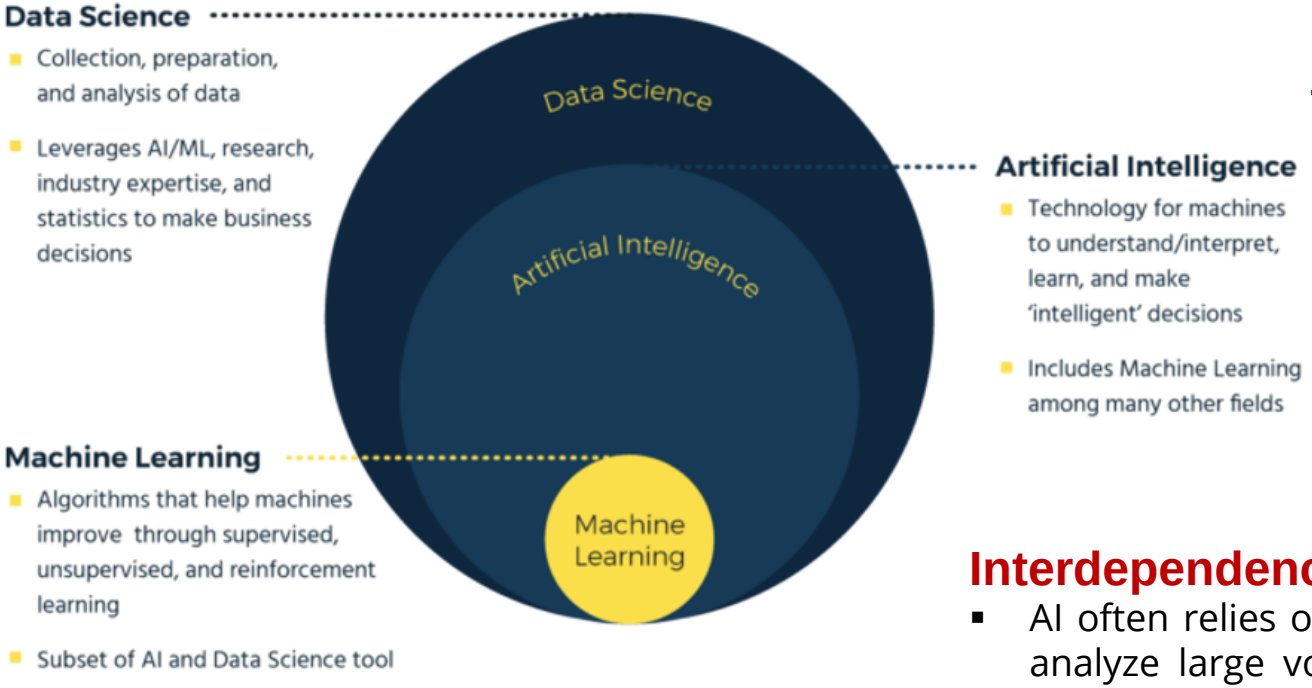
Traditional Programming VS Machine Learning?

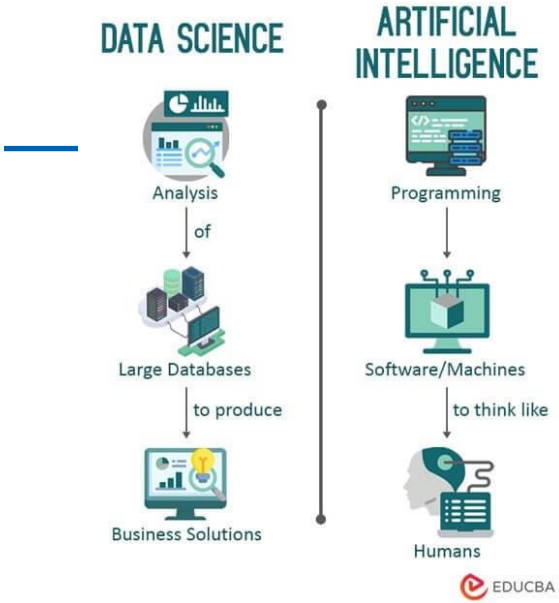
Basic Difference in ML and Traditional Programming?

Traditional Programming : We feed in DATA (Input) + PROGRAM (logic), run it on machine and get output.

Machine Learning : We feed in DATA(Input) + Output, run it on machine during training and the machine creates its own program(logic), which can be evaluated while testing.

Artificial Intelligence VS Data Science VS Machine Learning





Interdependence:

- AI often relies on Data Science to process and analyze large volumes of data, which is then used to improve AI systems.
- ML is a core component of both AI and Data Science, providing the techniques for learning from data and making predictions.

- Data Science deals with structured and unstructured data.
- Artificial Intelligence uses logic and decision trees.
- Machine Learning uses statistical models.

- Data science produces insights
- Machine learning produces predictions
- Artificial intelligence produces actions

AI VS ML VS Data Science

Artificial Intelligence (AI)

- AI is the broadest field, encompassing any technique that enables computers to mimic human intelligence.
- This includes reasoning, problem-solving, learning, and understanding natural language.

Applications: AI is used in various applications such as:

- Virtual Assistants: Siri, Alexa, and Google Assistant use AI to understand and respond to human queries.
- Autonomous Vehicles: Self-driving cars like those from Tesla use AI to navigate and make driving decisions.
- An AI system can analyze customer service interactions to understand and respond to customer queries efficiently.
- This includes recognizing speech, understanding the context, and generating appropriate responses.

AI VS ML VS Data Science

Data Science

Data Science involves extracting knowledge and insights from structured and unstructured data using various techniques from statistics, machine learning, data analysis, and domain expertise.

Applications: Data Science is applied in:

- **Healthcare: Predicting disease outbreaks and personalizing treatment plans.**
- **Finance: Fraud detection and risk management.**

Example: A data scientist might analyze large datasets of medical records to identify patterns and correlations that could lead to early diagnosis of diseases.

They use statistical models and visualization tools to interpret the data and present actionable insights.

AI VS ML VS Data Science

Machine Learning

Definition: ML is a subset of AI that involves training algorithms on data so they can learn and make predictions or decisions without being explicitly programmed for specific tasks.

Applications: ML is used in:

- **Recommendation Systems: Netflix and Amazon use ML algorithms to suggest movies or products based on user behavior.**
- **Spam Filtering: Email services use ML to identify and filter out spam messages.**

Example: An ML algorithm could be trained on historical sales data to predict future sales trends.
The model learns from patterns in the data and makes accurate predictions that help businesses plan inventory and marketing strategies.

AI VS ML VS Data Science

Real-World Example Integrating All Three:

Healthcare Diagnosis System:

- **Data Science: Collect and analyze patient data, including medical history, lab results, and genetic information.**
- **Machine Learning: Train models on this data to predict disease outcomes or suggest treatments.**
- **Artificial Intelligence: Implement an AI system that uses these ML models to assist doctors by providing diagnostic recommendations and treatment options in real time**

E-Commerce Personalization

- **Data Science: Collect and analyze user data such as browsing history, purchase history, and demographic information.**
- **Machine Learning: Train recommendation algorithms on this data to predict what products users might be interested in.**
- **Artificial Intelligence: Implement an AI system that uses these ML models to provide personalized product recommendations to users in real-time, enhance search results, and optimize marketing strategies.**

Traditional Approach - Spam Filtering

Spam typically uses words or phrases such as “4U,” “credit card,” “free,” and “amazing”

Solution

Write a detection algorithm for frequently appearing patterns in spams

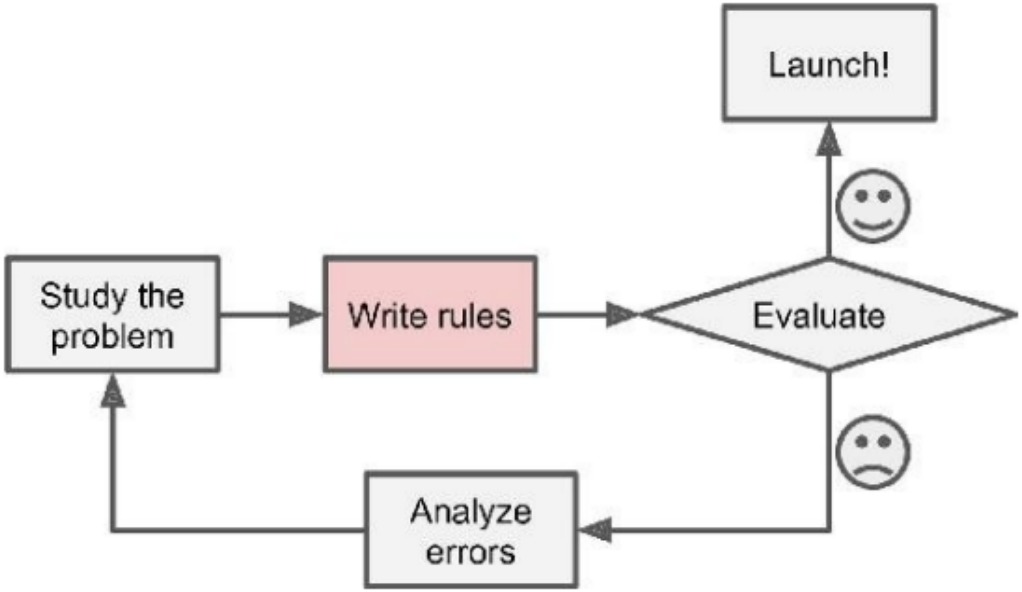
Test and update the detection rules until it is good enough.

Challenge

Detection algorithm likely

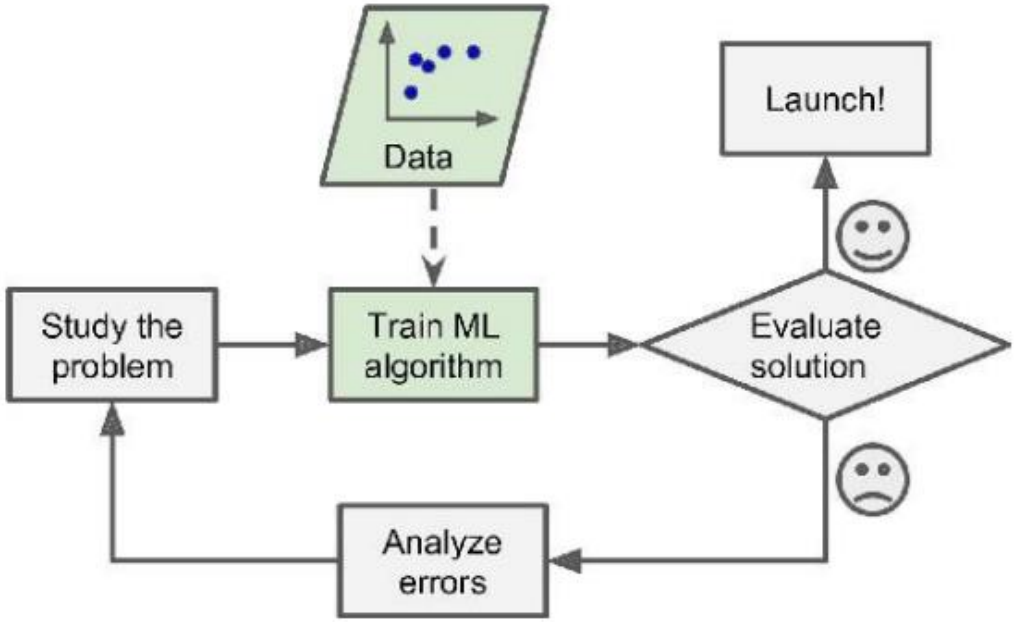
to be a long list of complex rules

hard to maintain.



ML Approach - Spam Filtering

Automatically learns phrases that are good predictors of spam by detecting unusually frequent patterns of words in spams compared to “ham”s



The program is much shorter, easier to maintain, and most likely more accurate.

ML Approach - Spam Filtering

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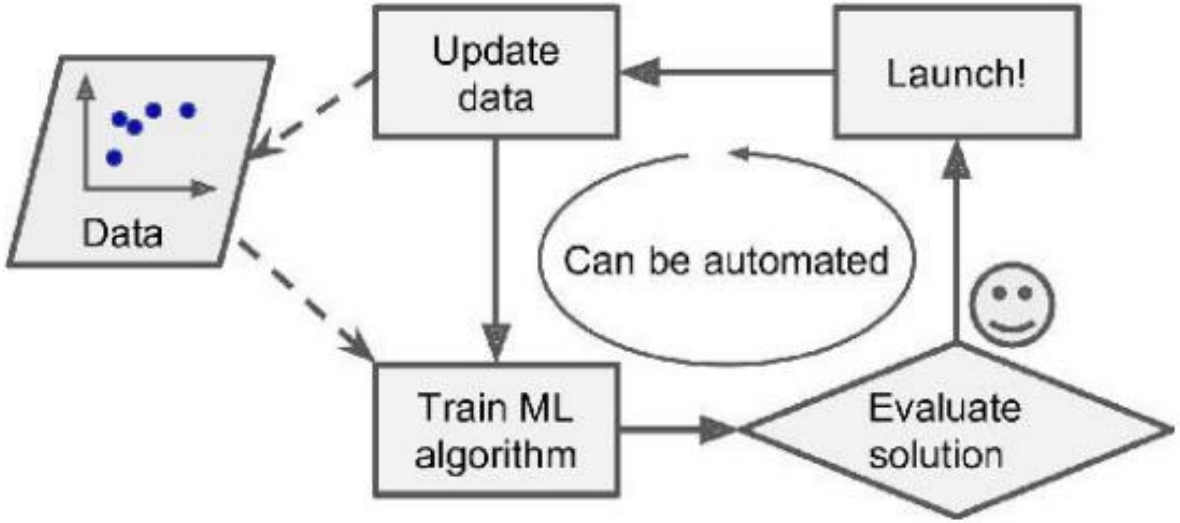


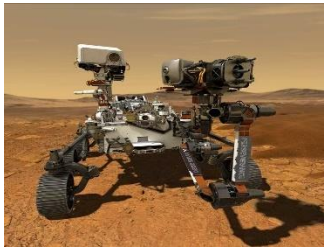
The program is much shorter, easier to maintain, and most likely more accurate.

Why ML

When Do We Use Machine Learning?

ML is used when:

- Human expertise does not exist (navigating on Mars)



- Humans can’t explain their expertise (Biometrics)

- Models must be customized (personalized medicine)

- Learning isn’t always useful:
  - There is no need to “learn” to calculate payroll

Need For Machine Learning?

- Some tasks cannot be defined well, except by examples.
- Relationships and correlations can be hidden within large amounts of data.
- Machine - Learning may be able to find these relationships.
- The amount of knowledge available about certain tasks might be too large for explicit encoding by humans (e.g., medical diagnostic).

- We generate around 2.5 quintillion bytes **(1 Quintillion is 10^18)** of data every single day!
- **Increase in Data Generation:** Due to excessive production of data, we need a method that can be used **to structure, analyze and draw useful insights from data**. ML uses data to solve problems and find solutions to the most complex tasks faced by organizations.
- **Improve Decision Making:** By making use of various algorithms, Machine Learning can be used to make **better business decisions**. For example, Machine Learning is used to forecast sales, predict downfalls in the stock market, identify risks and anomalies, etc.

Need For Machine Learning?

Fraud Detection in Financial Transactions

- Scenario: Banks and financial institutions need to detect fraudulent transactions to protect their customers and minimize losses.
- ML Solution: Supervised learning algorithms like Random Forests or Gradient Boosting can be trained on historical transaction data labeled as fraudulent or non-fraudulent. These models learn to recognize patterns and anomalies associated with fraud.
- Why ML is Suitable: ML models can process large volumes of data quickly, adapt to new fraud patterns over time, and provide real-time fraud detection by scoring each transaction for its likelihood of being fraudulent.

Predictive Maintenance in Manufacturing

- Scenario: Manufacturing companies want to predict equipment failures before they happen to reduce downtime and maintenance costs.
- ML Solution: Time series analysis and anomaly detection algorithms can analyze sensor data from machinery to predict when a piece of equipment is likely to fail.
- Why ML is Suitable: ML models can handle large amounts of sensor data, detect subtle patterns indicating potential failures, and provide timely alerts for maintenance, thus preventing unplanned downtime.

Need For Machine Learning?

Speech Recognition in Virtual Assistants

- **Scenario:** Virtual assistants like Siri, Alexa, and Google Assistant need to accurately recognize and respond to voice commands.
- **ML Solution:** Deep learning models such as Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) can be trained on large datasets of voice recordings to transcribe spoken language.
- **Why ML is Suitable:** ML techniques can handle variability in speech, accents, and background noise, providing robust and accurate speech recognition capabilities for virtual assistants.

Customer Segmentation in Marketing

- **Scenario:** Businesses want to segment their customers to target marketing campaigns more effectively.
- **ML Solution**: Clustering algorithms like K-Means or DBSCAN can analyze customer data based on behavior, demographics, and purchase history to identify distinct customer segments.
- **Why ML is Suitable:** ML techniques can handle high-dimensional data, uncover hidden patterns, and group customers into meaningful segments for personalized marketing strategies.

Limitations of ML

Example: Designing a New Product

- **ML Limitation:** An ML model can analyze data and generate designs based on existing patterns, but it might lack true creativity or the ability to think outside the box. It can optimize based on past data but struggle to create genuinely innovative solutions.
- **AI Solution:** AI systems that combine ML with generative models, reinforcement learning, and heuristic search algorithms can explore a vast space of possibilities and come up with novel designs. These systems can simulate various scenarios, learn from feedback, and iteratively improve their designs, achieving levels of creativity and innovation that ML alone cannot.

General Intelligence and Common Sense Reasoning

Example: Personal Assistants

- **ML Limitation:** ML models can handle specific tasks well, such as scheduling meetings or setting reminders. However, they often lack general intelligence and common sense reasoning needed to handle unexpected queries, understand nuanced human behavior, or perform tasks that require a broad understanding of the world.
- **AI Solution:** AI systems that incorporate a combination of ML, knowledge graphs, natural language understanding, and world models can better simulate human-like understanding and reasoning. They can draw on a wide range of knowledge and contextual information to perform tasks that require common sense and general intelligence.

Limitations of ML

Adaptive Learning and Personalization

Example: Personalized Education Platforms

- ML Limitation: ML models can recommend educational content based on past performance data, but they may struggle to adapt to new learning styles, identify gaps in knowledge, or personalize learning experiences in real-time.
- AI Solution: AI systems that integrate ML with cognitive architectures, adaptive learning algorithms, and student modeling can create highly personalized learning experiences. These systems can dynamically adjust to individual learning needs, provide tailored feedback, and evolve with the learner over time, offering a more holistic educational experience.

Understanding Context in Natural Language Processing (NLP)

Example: Conversational AI in Customer Support

- ML Limitation: An ML model, such as a basic NLP classifier, can be trained to recognize and respond to specific customer queries based on historical data. However, it may struggle with understanding the context, handling ambiguous language, or managing multi-turn conversations that require maintaining context across interactions.
- AI Solution: A comprehensive AI system, including techniques like deep learning, context-aware NLP, and reinforcement learning, can better understand and manage conversations. This involves not just recognizing patterns but also understanding the context, user intent, and previous interactions to provide coherent and contextually appropriate responses.

Test Yourself

Which of the following best describes the relationship between AI, Machine Learning, and Data Science?

A. Machine Learning is a subset of Data Science, and Data Science is a subset of AI.
B. AI is a subset of Machine Learning, and Machine Learning is a subset of Data Science.
C. Data Science is a subset of AI, and Machine Learning is a subset of Data Science.
D. Machine Learning is a subset of AI, and Data Science uses techniques from both AI and Machine Learning.

Answer: D. Machine Learning is a subset of AI, and Data Science uses techniques from both AI and Machine Learning.

Which scenario is least likely to be effectively solved using only Machine Learning techniques without broader AI integration?

A. Predicting stock prices based on historical data.
B. Recognizing objects in images for a photo tagging application.
C. Maintaining coherent and context-aware conversations in a chatbot over multiple interactions.
D. Recommending products to users based on their past purchase history.

Answer: C. Maintaining coherent and context-aware conversations in a chatbot over multiple interactions.

Test Yourself

Why might a data science project need to incorporate AI techniques beyond just machine learning models?

A. To analyze structured data efficiently.
B. To automate the cleaning and preprocessing of data.
C. To enhance the system's ability to reason and make complex decisions based on context.
D. To create visualizations for data insights.
Answer: C. To enhance the system's ability to reason and make complex decisions based on context.

Which of the following limitations of Machine Learning is addressed by incorporating broader AI techniques like symbolic reasoning?

A. Difficulty in processing large datasets.
B. Lack of understanding and reasoning about unseen scenarios.
C. Slow training times for complex models.
D. Inability to perform simple arithmetic operations.

Answer: B. Lack of understanding and reasoning about unseen scenarios.

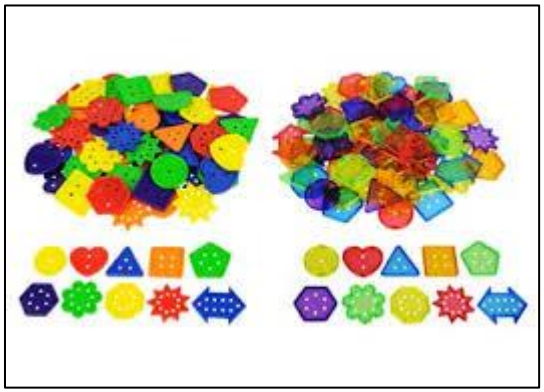
Technique	Description	Domains
Machine Learning (ML)	Training algorithms on data to make predictions or decisions without being explicitly programmed	Finance (fraud detection), E-commerce (recommendation systems), Healthcare (disease prediction)
Deep Learning	Neural networks with many layers that can learn from large amounts of data	Computer Vision (image recognition), NLP (language translation), Autonomous Vehicles (object detection)
Natural Language Processing (NLP)	Enabling machines to understand, interpret, and generate human language	Customer Service (chatbots), Content Analysis (sentiment analysis), Translation Services (language translation)
Computer Vision	Enabling machines to interpret and understand visual information from the world	Healthcare (medical imaging), Security (facial recognition), Automotive (autonomous driving)
Expert Systems	AI programs that emulate human expert decision-making using a set of rules and a knowledge base	Medical Diagnosis (decision support systems), Finance (investment analysis), Customer Support (troubleshooting)
Robotics	The design and use of robots to perform tasks, often using AI to enhance capabilities	Manufacturing (automation), Healthcare (surgical robots), Space Exploration (autonomous rovers)
Reinforcement Learning	An agent learns to make decisions by performing actions and receiving feedback (rewards or penalties)	Gaming (game AI), Robotics (autonomous robots), Finance (trading algorithms)
Genetic Algorithms	Optimization algorithms inspired by natural selection, evolving solutions over iterations	Engineering (design optimization), Finance (portfolio optimization), Artificial Life (evolving strategies)
Knowledge Representation and Reasoning (KRR)	Representing information about the world for complex task solving	Healthcare (clinical decision support), Legal (legal reasoning), AI Planning (task scheduling)
Fuzzy Logic	Reasoning that is approximate rather than fixed and exact, handling uncertainty and vagueness	Control Systems (temperature control), Consumer Electronics (washing machines), Automotive (anti-lock braking systems)
Bayesian Networks	Probabilistic graphical models representing variables and their conditional dependencies	Healthcare (diagnostic systems), Finance (risk assessment), Engineering (fault diagnosis)
Semantic Web	Making web data machine-readable for complex human requests based on their meaning	Information Retrieval (semantic search), Data Integration (linked data), Personal Assistants (contextual understanding)
Cognitive Computing	Systems that simulate human thought processes to improve human decision-making	Healthcare (personalized treatment plans), Customer Support (virtual agents), Education (adaptive learning)
Swarm Intelligence	Optimization techniques inspired by the collective behavior of decentralized, self-organized systems	Robotics (multi-robot systems), Telecommunications (network routing), Logistics (fleet management)
Neural Networks	Computational models inspired by the human brain, with interconnected nodes (neurons)	Speech Recognition (voice assistants), Finance (market prediction), Healthcare (diagnostic imaging)

Types of ML

Introduction to Machine Learning

Types : Inputs: Based on level of supervision

Feedback	No Feedback	Delayed Feedback (rewards/penalty)
Supervised	Unsupervised	Reinforcement



Machine Learning - Examples

Objective: Employability Prediction

Features / Attributes / Predictors

- ✓ CGPA
- ✓ Communication Skills
- ✓ Aptitude
- ✓ Programming Skills

S.No.	CGPA	Communication Skills	Aptitude	Programming Skills	Job Offered?
1	9.1	Average	Good	Excellent	Yes
2	8.4	Good	Good	Good	Yes
3	8.3	Poor	Average	Average	No
4	7.1	Average	Good	Average	No
5	8.2	Good	Excellent	Excellent	No

Machine Learning - Examples

Objective : Predicting price of a used car

Features / Attributes / Predictors

- ✓ Brand
- ✓ Year (Mfg)
- ✓ Engine Capacity
- ✓ Mileage
- ✓ Distance travelled
- ✓ Cab?

S.No	Brand	Year (Mfg)	Engine Capacity	Mileage	Distance travelled	Cab?	Price (in Rs.)
1.	Honda City ZX	2008	1100	10.5	45000	N	3,50,000
2							
3							
4							

Machine Learning - Examples

Objective: Market Segmentation Study

Features / Attributes / Predictors

- ✓ Family income
- ✓ # of visits in a month
- ✓ Average money spent in a month
- ✓ Zip code

Customers for a retailer may fall into

- ✓ two groups say big spenders and low spenders
- ✓ three groups say big spenders, medium spenders and low spenders
- ✓ Four groups, ….

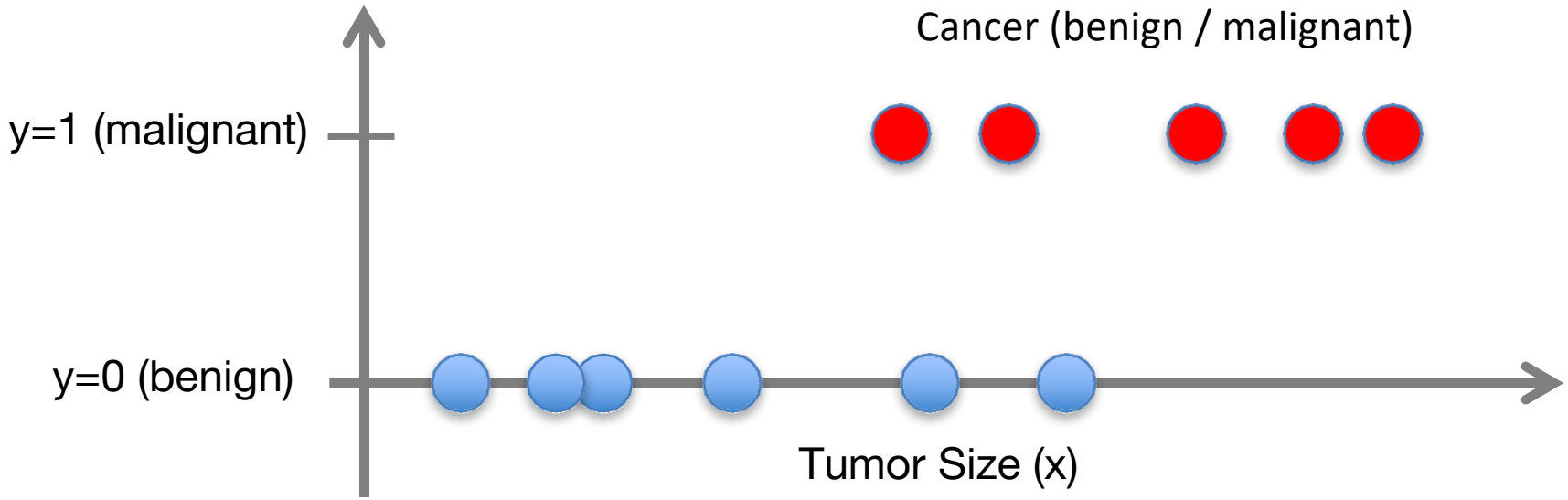
S.No.	Zip Code	Family Income	# of visits in a month	Average Money Spent in a month
1	500078	11,50,000	4	8,000

Introduction to Machine Learning

Supervised Learning: Classification

GOAL : Previously unseen records should be assigned a class as accurately as possible.

- Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
  - y is categorical

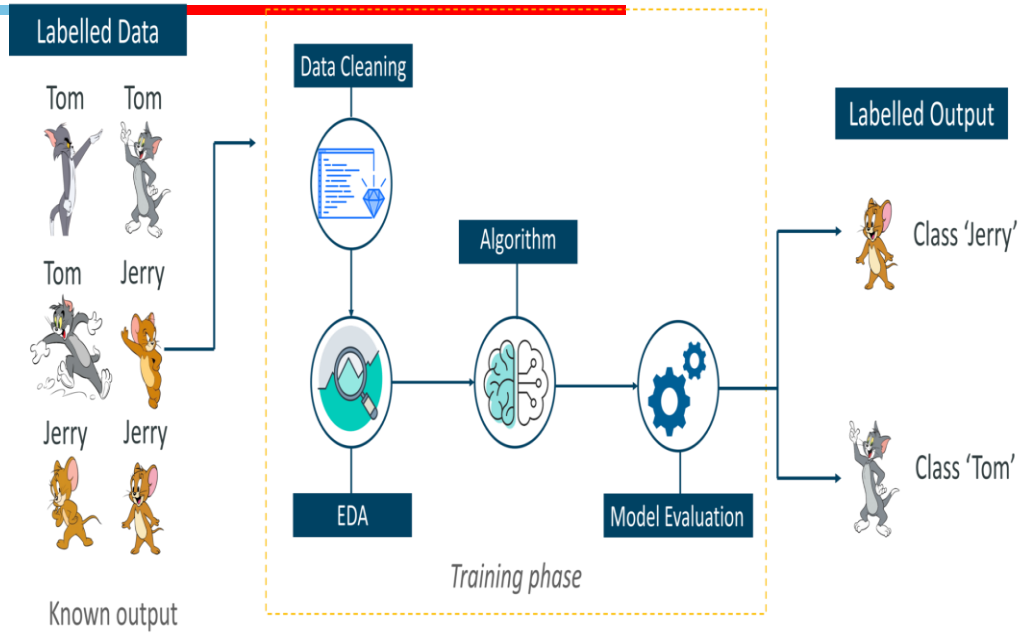


Slide Credit: Eric Eaton

Introduction to Machine Learning

Supervised Learning

- Supervised learning is a technique in which we train the machine using data which is well labeled.
- Applications in which the training data comprises examples of the input vectors along with their corresponding target vectors are known as supervised learning problems.

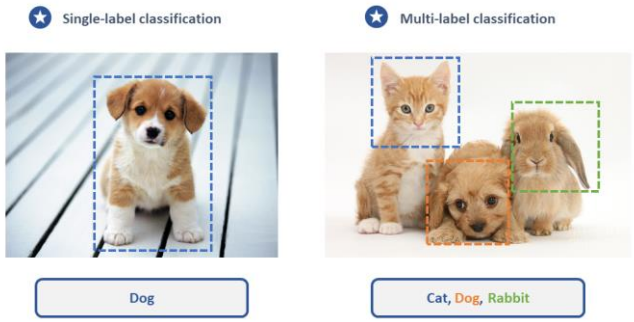


- The training data provided to the machines work as the supervisor that teaches the machines to predict the output correctly.
- It applies the same concept as a student learns in the supervision of the teacher.
- As input data is fed into the model, it adjusts its weights until the model has been fitted appropriately.
- Supervised machine learning algorithms define models that capture relationships among data.

Introduction to Machine Learning

Classification Applications

- Google Image Classification
- Face recognition system
- Spam filters – Specific Controls
- Document tagging
- Fraud detection

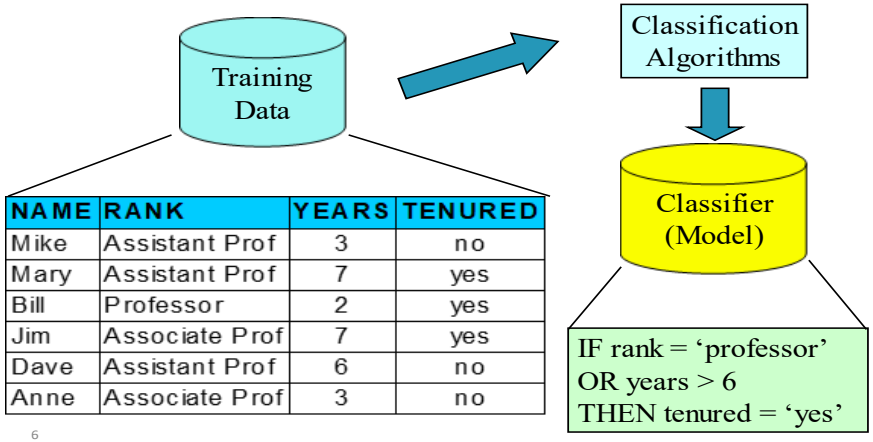


Supervised Learning Techniques / Algorithms

- Linear Regression
- Logistic Regression
- Naïve Bayes Classifiers
- Support Vector Machines (SVMs)
- Decision Trees and Random Forests
- Neural networks

Working of Supervised Learning Method

Process (1): Model Construction



NAME	RANK	YEARS	TENURED
Mike	Assistant Prof	3	no
Mary	Assistant Prof	7	yes
Bill	Professor	2	yes
Jim	Associate Prof	7	yes
Dave	Assistant Prof	6	no
Anne	Associate Prof	3	no

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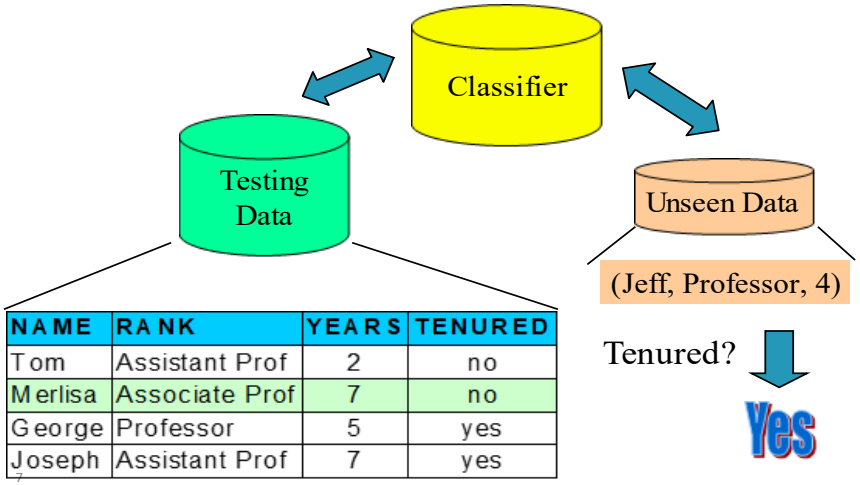
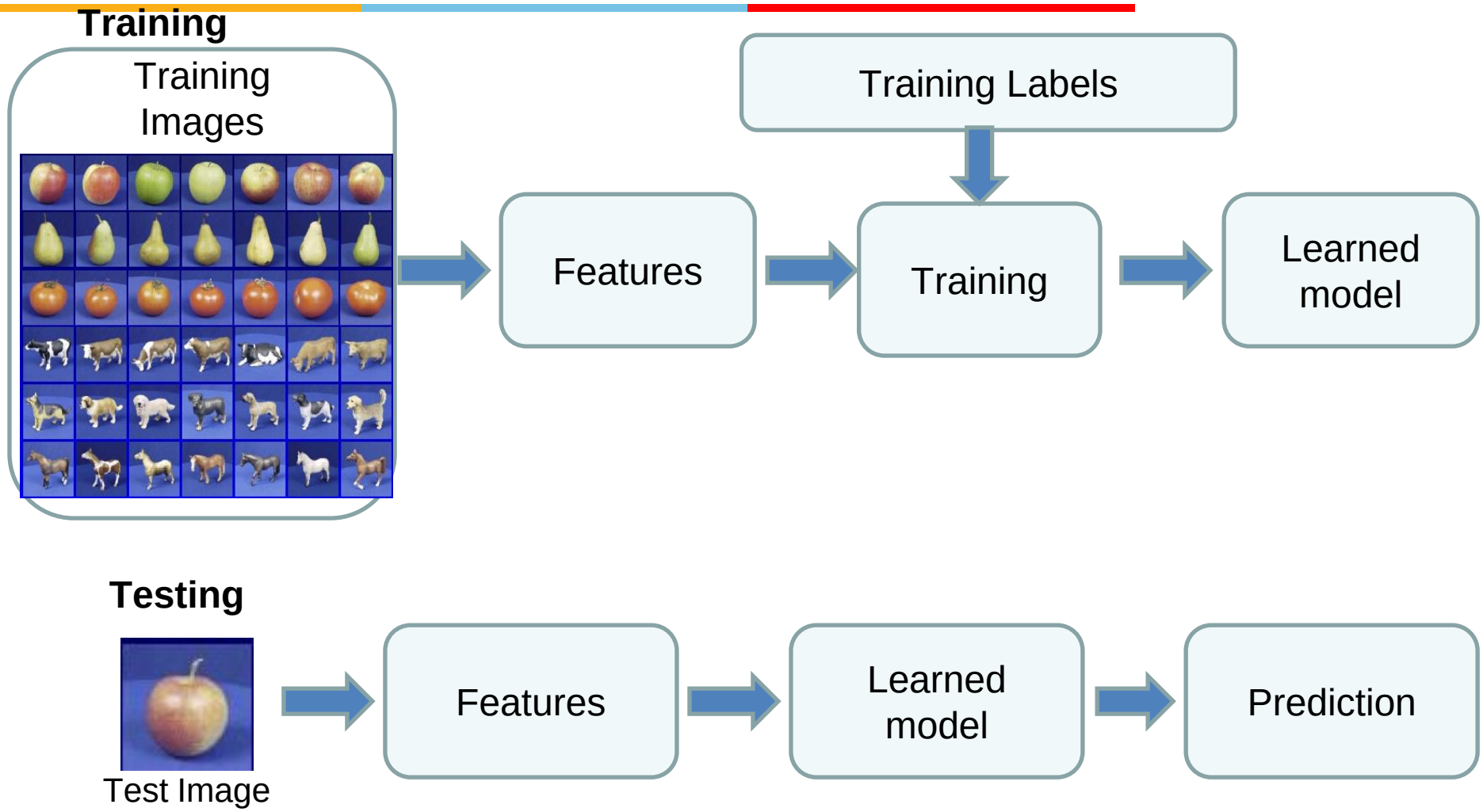
Process (2): Using the Model in Prediction



NAME	RANK	YEARS	TENURED
Tom	Assistant Prof	2	no
Merlisa	Associate Prof	7	no
George	Professor	5	yes
Joseph	Assistant Prof	7	yes

7

A Typical Supervised Learning Workflow (for Classification)



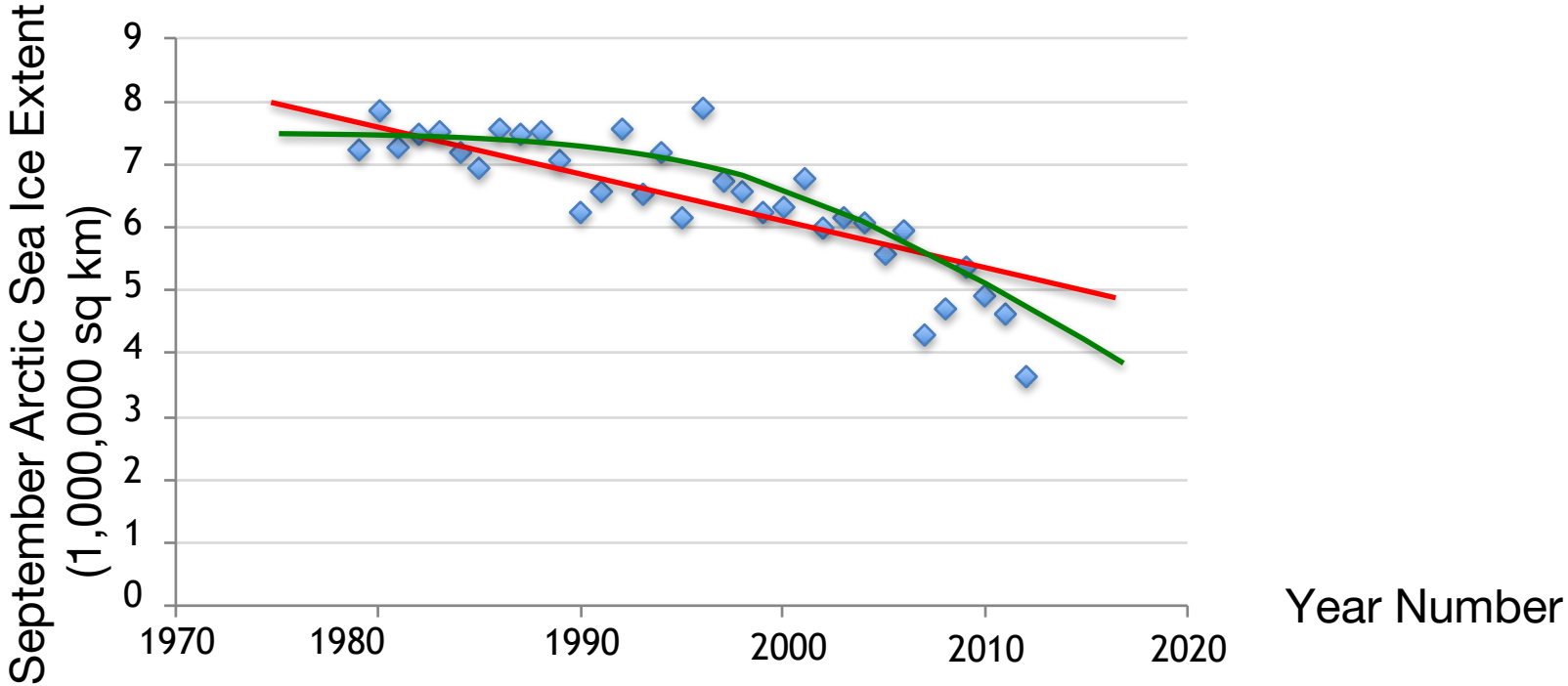
Slide credit: D. Hoiem and L. Lazebnik

Introduction to Machine Learning

Supervised Learning: Regression

GOAL : Previously unseen records should be assigned a value as accurately as possible.

- Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
  - y is real-valued



Data from G. Witt. Journal of Statistics Education, Volume 21, Number 1 (2013) Slide Credit: Eric Eaton

Regression

$$y = f(\mathbf{x})$$

output prediction function features

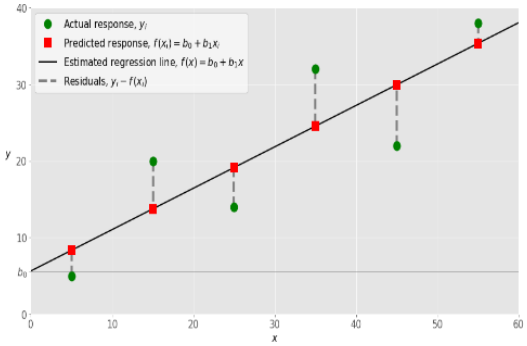
- **Training:** given a *training set* of labeled examples $\{(\mathbf{x}_1,y_1), \ldots, (\mathbf{x}_N,y_N)\}$, estimate the prediction function f by minimizing the prediction error on the training set
- **Testing:** apply f to a never before seen *test example* $\mathbf{x}$ and output the predicted value $y = f(\mathbf{x})$

Slide credit: L. Lazebnik

Regression

- Regression analysis is used to find trends in data.
- Typically, you need regression to answer whether and how some phenomenon influences the other or how several variables are related.
- It is used for the prediction of continuous variables, such as Weather forecasting, Market Trends, etc.
- For example, you can use it to determine *if* and *to what extent* the experience or gender impact salaries.
- For example, if you've been putting on weight over the last few years, it can predict how much you'll weigh in ten years time if you continue to put on weight at the same rate
- Essentially, regression is the "best guess" at using a set of data to make some kind of prediction.
- It's fitting a set of points to a graph.

The simple linear model is expressed using the following equation: **Y = a + bX + ϵ**
Where:
Y – Dependent variable
X – Independent (explanatory) variable
a – Intercept
b – Slope
ϵ – Residual (error)

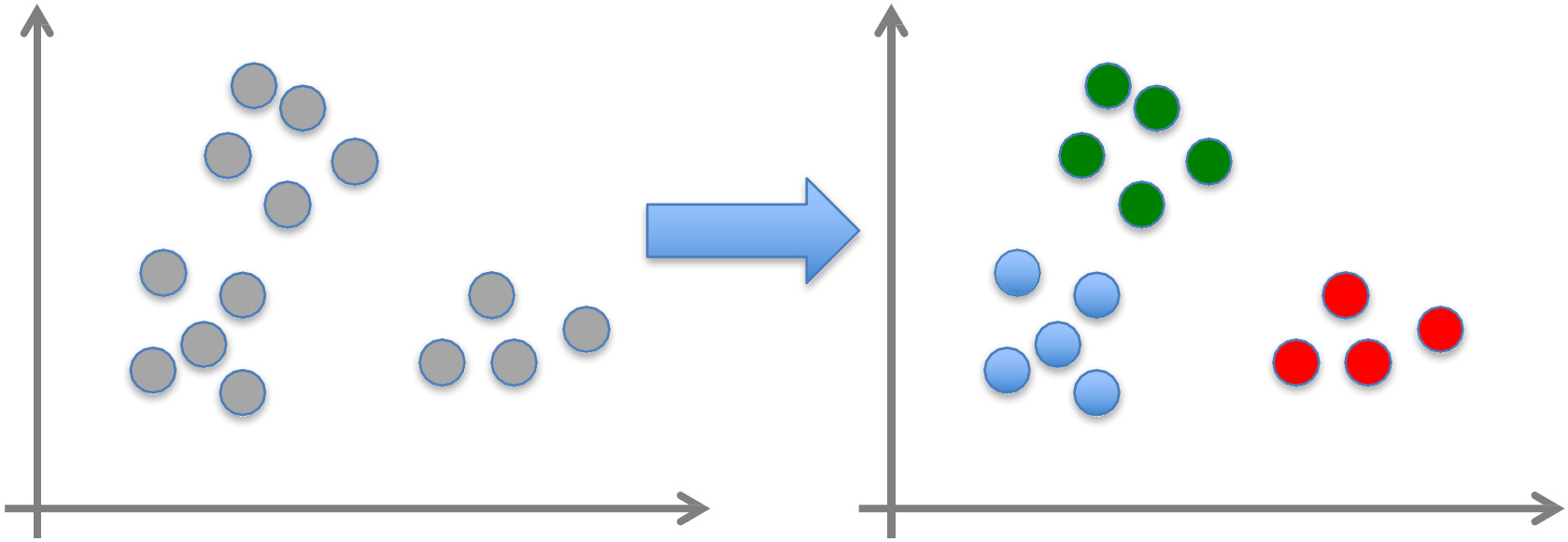


Introduction to Machine Learning

Unsupervised Learning

GOAL : Intra cluster distances are minimized and inter cluster distances are maximized

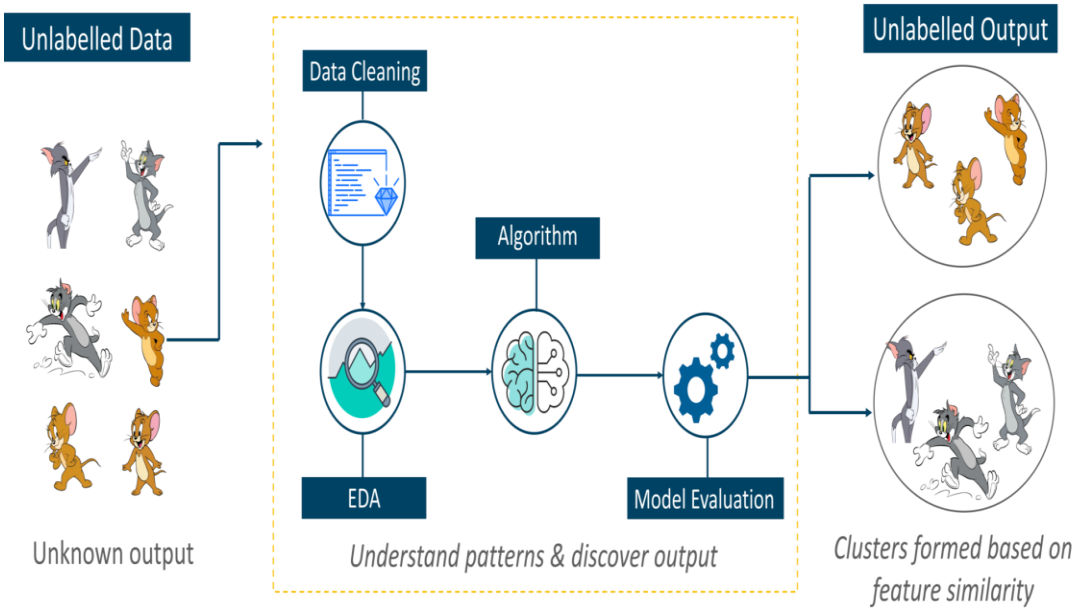
- Given $x_1, x_2, \ldots, x_n$ (without labels)
- Output hidden structure behind the x's
  - e.g., clustering



Slide Credit: Eric Eaton

Introduction to Machine Learning

Unsupervised Learning



For example, it identifies prominent features of Tom such as pointy ears, bigger size, etc, to understand that this image is of type 1. Similarly, it finds such features in Jerry and knows that this image is of type 2. Therefore, it classifies the images into two different classes without knowing who Tom is or Jerry is.

Slide Credit: Eric Eaton

Introduction to Machine Learning

Unsupervised Learning-Example

Let's, take the case of a baby and her family dog.

She knows and identifies this dog. Few weeks later a family friend brings along a dog and tries to play with the baby.

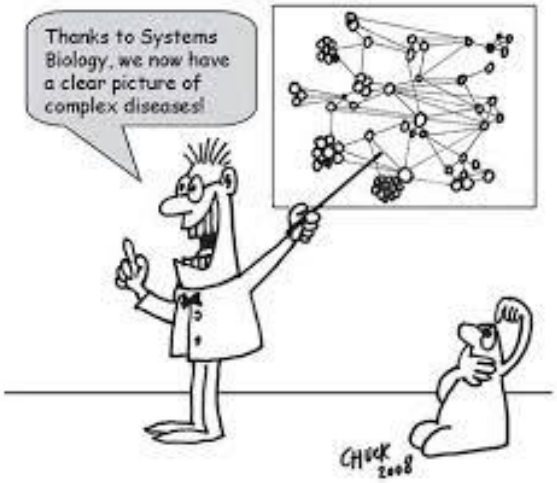
Baby has not seen this dog earlier. But it recognizes many features (2 ears, eyes, walking on 4 legs) are like her pet dog. She identifies the new animal as a dog. This is unsupervised learning, where you are not taught but you learn from the data (in this case data about a dog.)

Had this been supervised learning, the family friend would have told the baby that it's a dog.

Introduction to Machine Learning

Unsupervised Learning Applications

- Personalized recommendation system
- Targeted marketing
- Spam Filters
- Content Management – News hosted in Web
- Campaigning



Unsupervised Learning Techniques

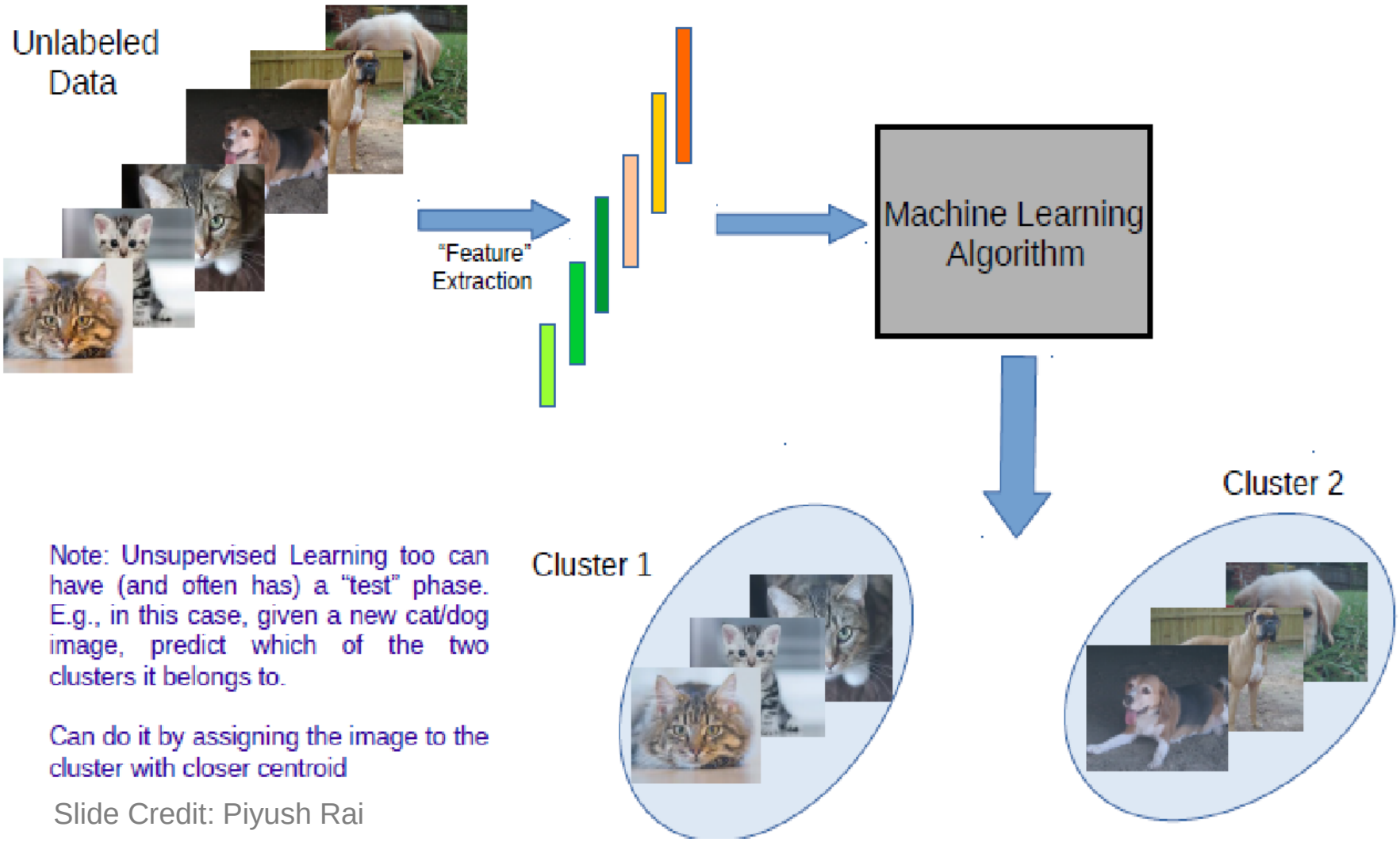
Clustering

- k-Means
- Hierarchical Cluster Analysis
- Expectation Maximization

Visualization and dimensionality reduction

- Principal Component Analysis (PCA)
- Kernel PCA
- Locally-Linear Embedding (LLE)
- t-distributed Stochastic Neighbor Embedding (t-SNE)

A Typical Unsupervised Learning Workflow (for Clustering)



Note: Unsupervised Learning too can have (and often has) a "test" phase. E.g., in this case, given a new cat/dog image, predict which of the two clusters it belongs to.

Can do it by assigning the image to the cluster with closer centroid

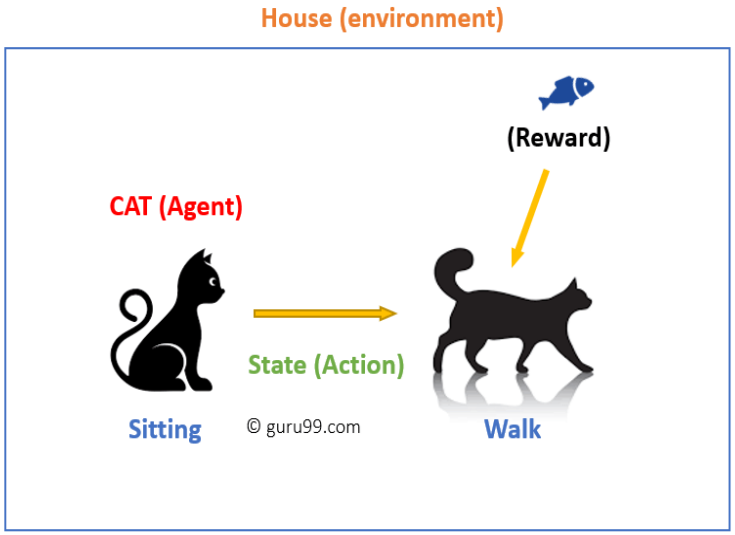
Slide Credit: Piyush Rai

Reinforcement Learning

- Reinforcement Learning is a part of Machine learning where an agent is put in an environment and he learns to behave in this environment by performing certain actions and observing the rewards which it gets from those actions.
- Imagine that you were dropped off at an isolated island! What would you do?
- As time passes by, you will learn how to live on the island. You will explore the environment, understand the climate condition, the type of food that grows there, the dangers of the island, etc. This is exactly how Reinforcement Learning works,
- it involves an **Agent** (**you, stuck on the island**) that is put in an unknown environment (island), where he must learn by observing and performing actions that result in rewards.
- Reinforcement Learning is mainly used in **advanced Machine Learning areas** such as self-driving cars, AplhaGo, etc.

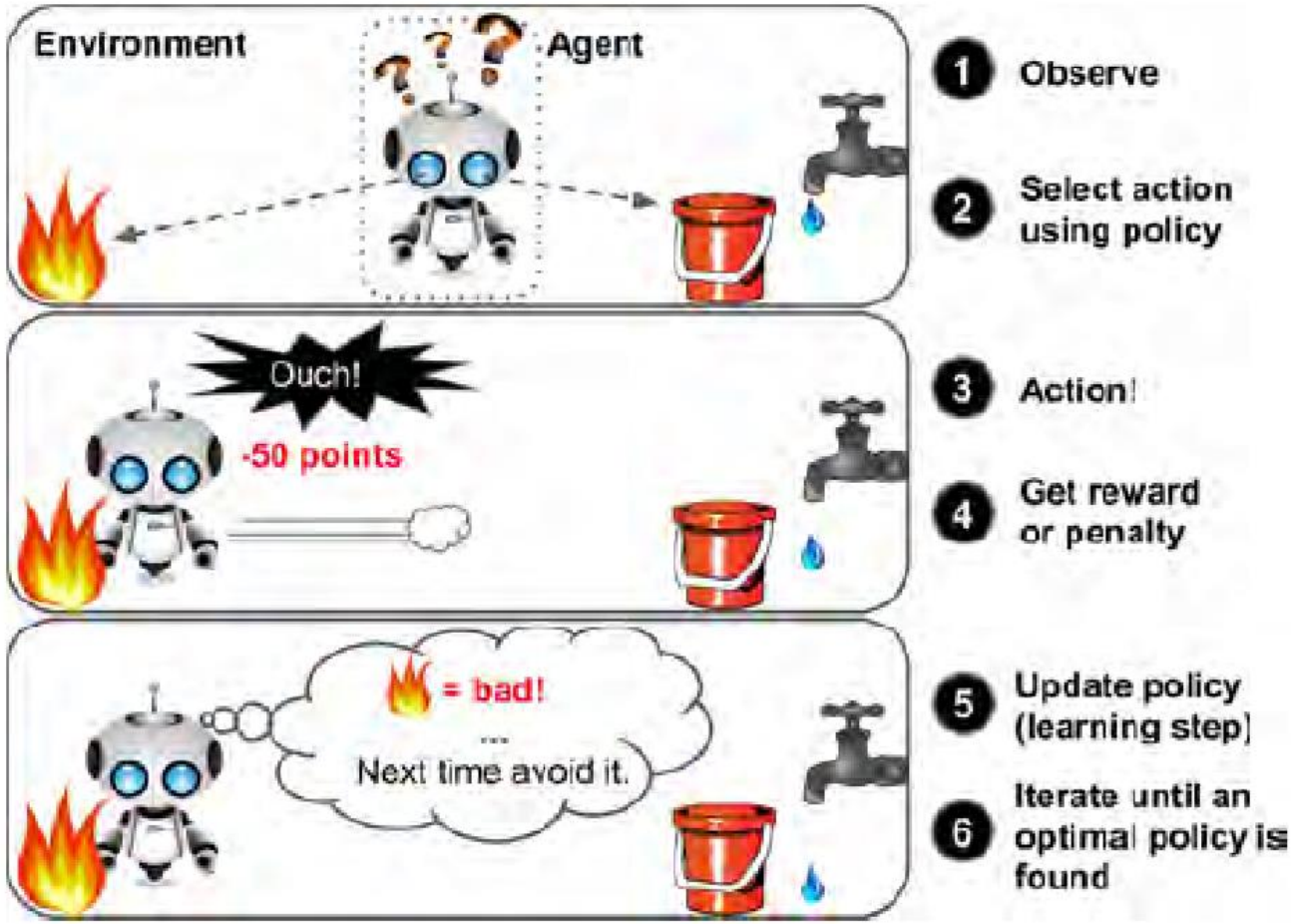
How Reinforcement Learning works?

- Consider the scenario of teaching new tricks to your cat
- As cat doesn't understand English or any other human language, we can't tell her directly what to do. Instead, we follow a different strategy.
- We emulate a situation, and the cat tries to respond in many different ways. If the cat's response is the desired way, we will give her fish.
- Now whenever the cat is exposed to the same situation, the cat executes a similar action with even more enthusiastically in expectation of getting more reward(food).
- That's like learning that cat gets from "what to do" from positive experiences.
- At the same time, the cat also learns what not do when faced with negative experiences.



- **There is no supervisor, only a real number or reward signal**
- **Sequential decision making**
- **Time plays a crucial role in Reinforcement problems**
- **Feedback is always delayed, not instantaneous**
- **Agent's actions determine the subsequent data it receives**

Reinforcement Learning



Some Examples of ML Applications

Traffic Alerts (Maps) : **Google Maps** is probably **the** app we use whenever we go out and require assistance in directions and traffic. "***Despite the Heavy Traffic, you are on the fastest route***". But, **How does it know that?**

▪Everyone using maps is providing their location, average speed, the route in which they are traveling which in turn helps Google collect massive Data about the traffic, which makes them predict the upcoming traffic and adjust your route according to it.

Social Media (Facebook) :One of the most common applications of Machine Learning is **Automatic Friend Tagging Suggestions** in Facebook or any other social media platform. Facebook uses **face detection** and **Image recognition** to automatically find the face of the person which matches it's Database and hence suggests us to tag that person based on DeepFace.

Transportation and Commuting (Uber) : If you have used an app to book a cab, you are already using Machine Learning to an extent. It provides a personalized application which is unique to you. Automatically detects your **location** and provides options to either go home or office or any other frequent place based on your **History and Patterns.**

Image courtesy: https://towardsdatascience.com/types-of-machine-learning-algorithms-you-should-know-953a08248861

Some Examples of ML Applications

- ***Amazon's Alexa:*** The infamous Alexa, which is based on Natural Language Processing and Machine Learning is an advanced level Virtual Assistant that does more than just play songs on your playlist. It can book you an Uber, connect with the other IoT devices at home, track your health, etc.
- ***Google's Spam Filter:*** Gmail makes use of Machine Learning to filter out spam messages. It uses Machine Learning algorithms and Natural Language Processing to analyze emails in real-time and classify them as either spam or non-spam.
- ***Products Recommendations:*** Suppose you check an item on Amazon, but you do not buy it then and there. But the next day, you're watching videos on YouTube and suddenly you see an ad for the same item. You switch to Facebook, there also you see the same ad. So how does this happen?
- ***Self Driving Cars*** **:** Machine Learning plays a very important role in Self Driving Cars and I'm sure you guys might have heard about **Tesla**. The leader in this business and their current ***Artificial Intelligence*** is driven by hardware manufacturer **NVIDIA**, which is based on Unsupervised Learning Algorithm. they didn't train their model to detect people or any object as such. The model works on ***Deep Learning*** and it crowdsources data from all of its vehicles and its drivers.

Image courtesy: https://towardsdatascience.com/types-of-machine-learning-algorithms-you-should-know-953a08248861

Summary : Types of Learning

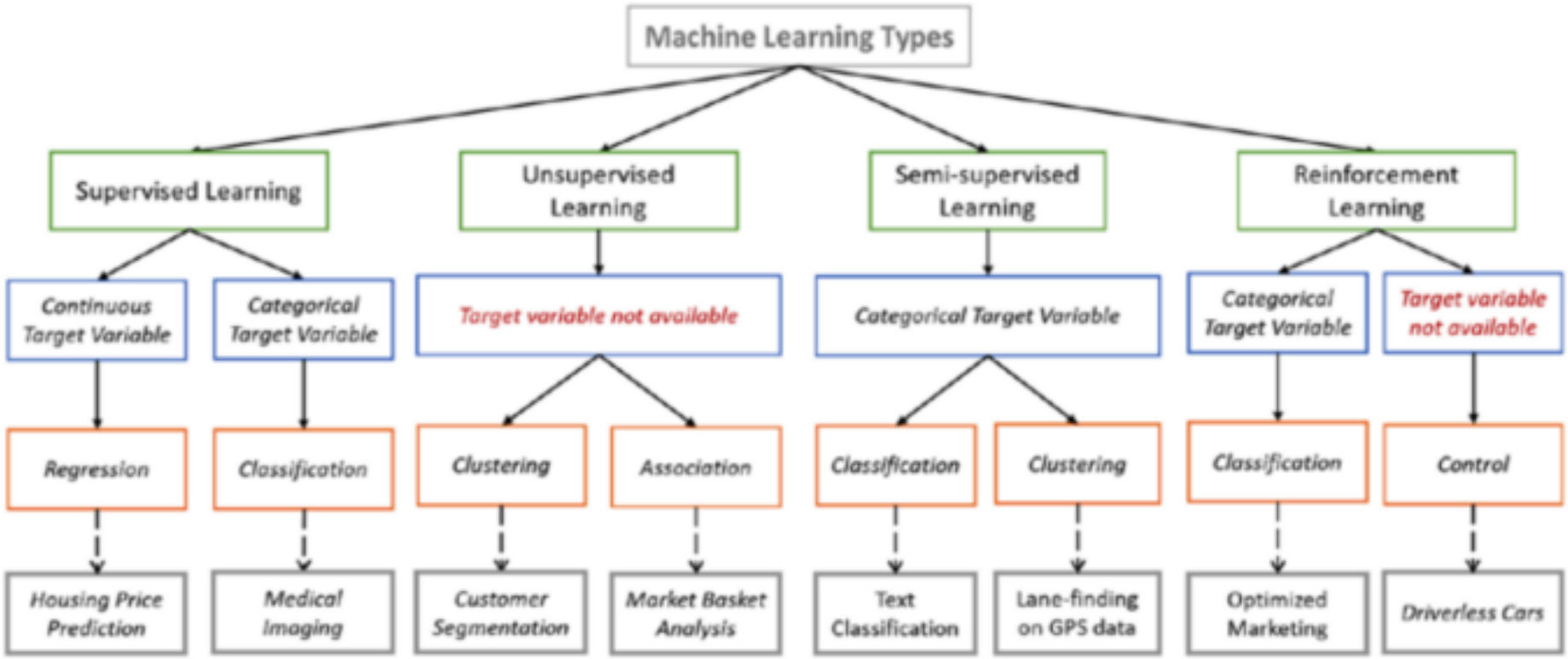


Image courtesy: https://towardsdatascience.com/types-of-machine-learning-algorithms-you-should-know-953a08248861

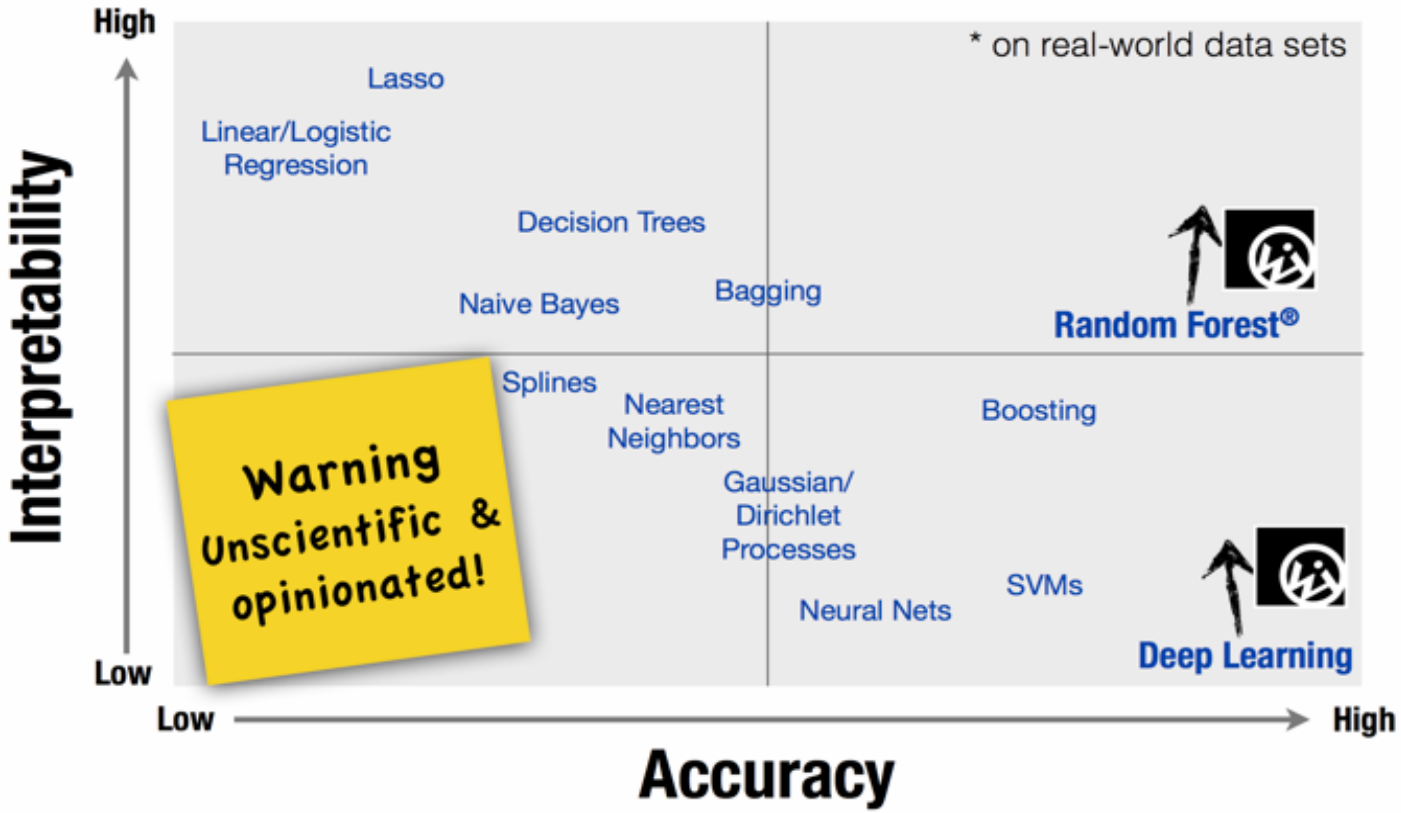
Open source ML programming tools

	Platform		Algorithms or Features
Scikit Learn	Linux, Mac OS, Windows	Python, C, C++	Classification, Regression, Clustering Preprocessing, Model Selection Dimensionality reduction.
PyTorch	Linux, Mac OS, Windows	Python, C++	Autograd Module, Optimization Module NN Module
TensorFlow	Linux, Mac OS, Windows	Python, C++	Provides a library for dataflow programming.
Weka	Linux, Mac OS, Windows	Java	Data preparation, Classification Regression, Clustering, Visualization Association rules mining

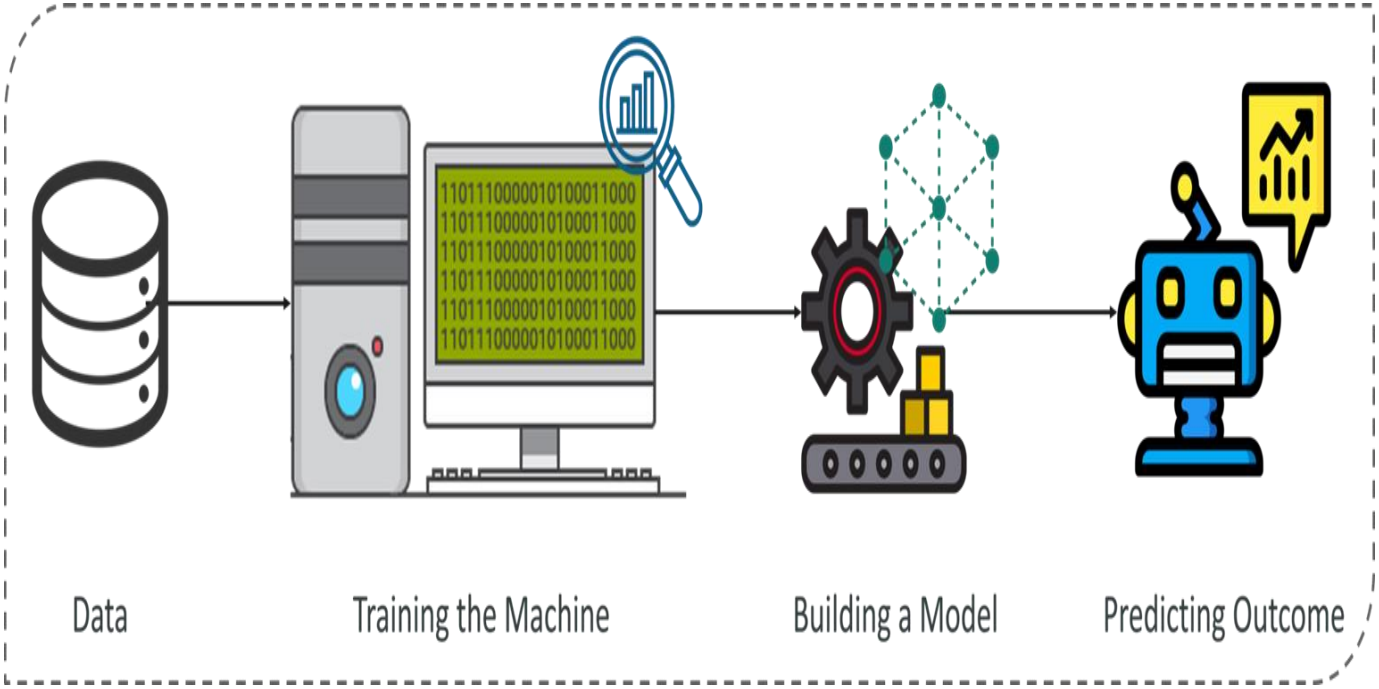
Open source ML programming tools

Colab	Cloud Service	-	Supports libraries of PyTorch, Keras, TensorFlow, and OpenCV
Apache Mahout	Cross-platform	Java Scala	Preprocessors,Regression Clustering, Recommenders Distributed Linear Algebra.
Accors.Net	Cross-platform	C#	Classification, Regression, Distribution Clustering, Hypothesis Tests & Kernel Methods, Image, Audio & Signal & Vision
Shogun	Windows Linux, UNIX Mac OS	C++	Regression, Classification, Clustering Support vector machines. Dimensionality reduction, Online learning etc.
Keras.io	Cross-platform	Python	API for neural networks

ML Algorithmic Trade-Off



ML workflow


Data
Training the Machine
Building a Model
Predicting Outcome

ML workflow

1. Should I use ML on this problem?
   - Is there a pattern to detect?
   - Can I solve it analytically?
   - Do I have data?
2. Gather and organize data.
3. Preprocessing, cleaning, visualizing.
4. Choosing a model, loss, regularization, ...
5. Optimization
6. Hyper parameter search.
7. Analyze performance and mistakes, and iterate back to step 5 (or 3)

Example: Car Price prediction based on Mileage

X Mileage	Y Car Price	H0	Error L1
9.8	10.48	9	2
9.12	1.75	8	7
9.5	6.95	9	3
10	2.51	9	7

- Define the Objective
- Data Gathering: survey, Past Purchase data
- Data Preprocessing
  - training set; test set
  - representation of input features; output
- Exploratory Data Analysis
- Choose form of model: linear regression
- System's performance evaluation: objective function
- Optimize performance by setting appropriate parameters: Optimization
- Evaluate on test set: generalization

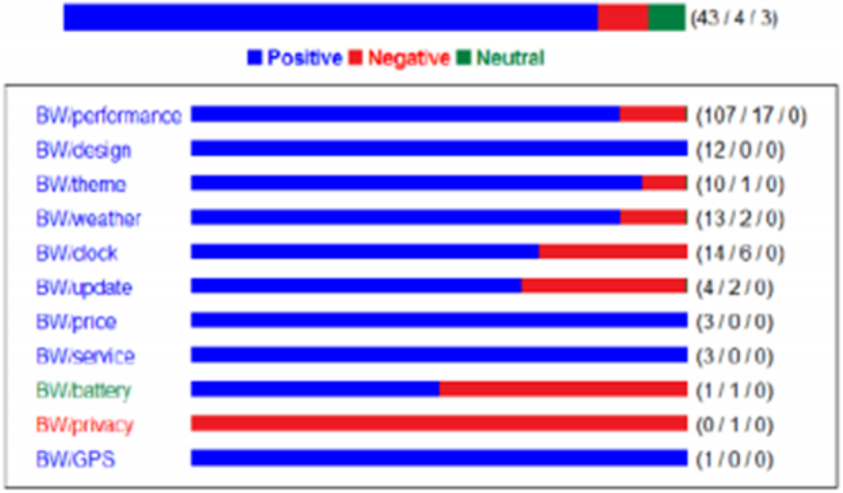
References

- Chapter 1 – Machine Learning, Tom Mitchell
- Chapter 1, 2 – Introduction to Machine Learning, 2nd edition, Ethem Alpaydin
- Chapter 1 - Pattern Recognition & Machine Learning Christopher M. Bhishop
- http://www.cs.princeton.edu/courses/archive/spr08/cos511/ [Web]
- https://www.softwaretestinghelp.com/machine-learning-tools/

Thank you !

Introduction to Machine Learning

Common Use cases - Security & Transaction Domain



Sentiment analysis on Product review of Mobile phone

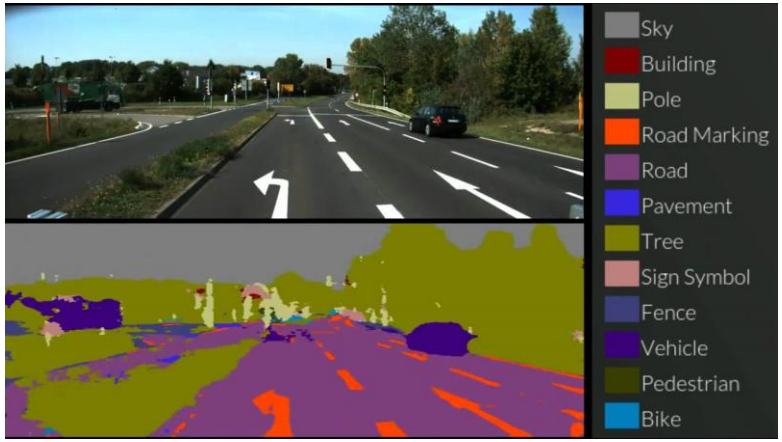
- Self Driving Cars
- Fraud Detection in Banking
- Email Filtering
- Dynamic Pricing in Travel

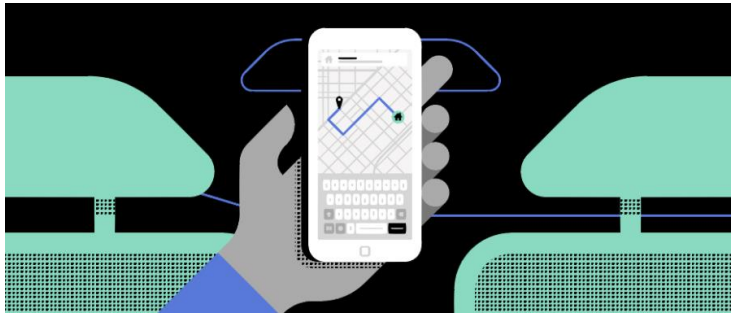
Derived Applications:

> Cyber Security

> Video Surveillance

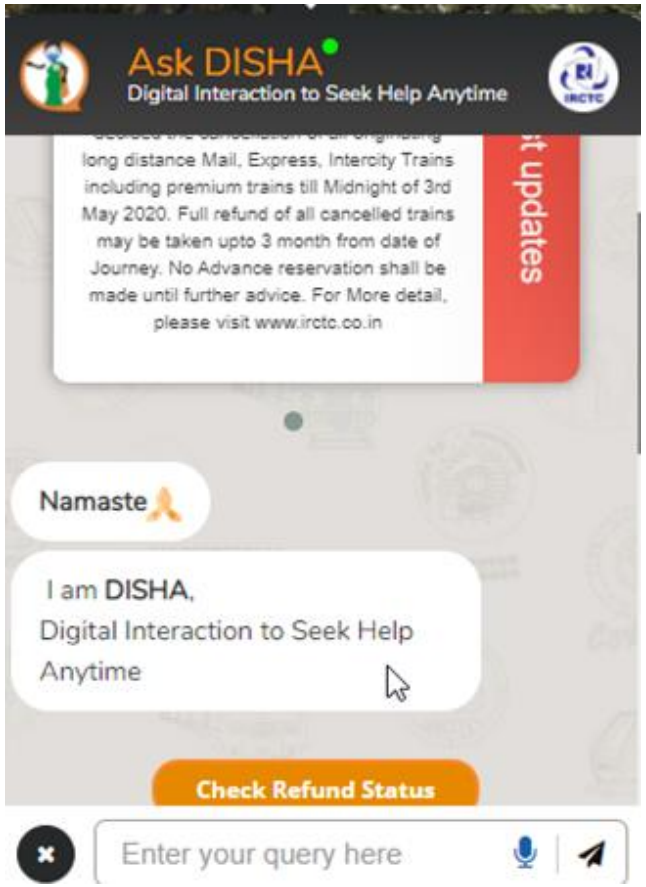
> Object Detection

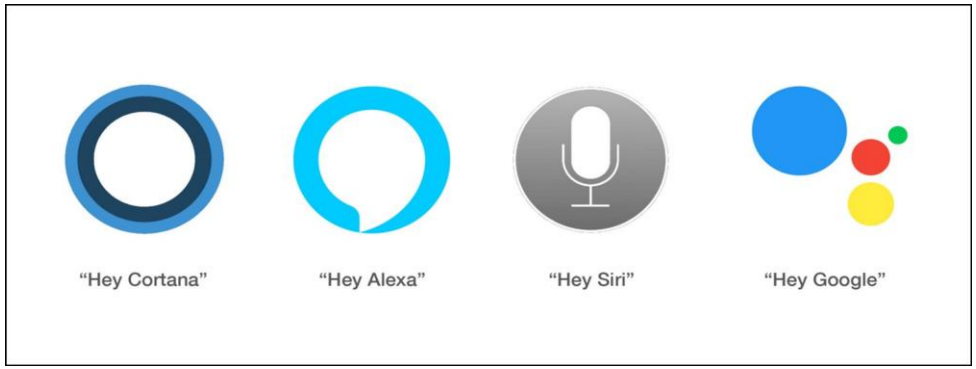




Introduction to Machine Learning

Common Use cases - Customer Support Systems





- Apple's Siri
- Google Assistant
- Amazon's Alexa
- Google Duplex
- Microsoft's Cortana
- Samsung's Bixby

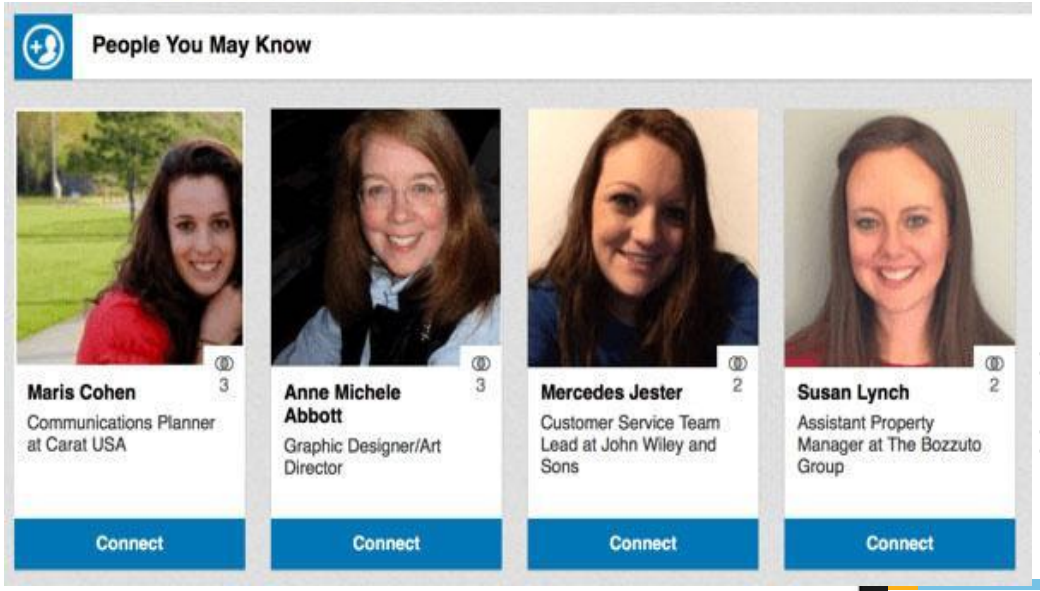
Derived Applications:

> Customer Support Query (Voice vs Text)

> Chatbots

Introduction to Machine Learning

Common Use cases - Recommendation Engines



- E-commerce sites like Amazon and Flipkart
- Book sites like Goodreads
- Movie services like IMDb and Netflix
- Hospitality sites like MakeMyTrip, Booking.com, etc.
- Retail services like StitchFix
- Food aggregators like Zomato and Uber Eats

Derived Applications:

> Personalized Marketing

> Personalized Banking

Data Mining Vs Machine Learning

- Data mining is designed to extract the rules from large quantities of data,
- while machine learning teaches a computer how to learn and comprehend the given parameters.
- Or to put it another way, data mining is simply a method of researching to determine a particular outcome based on the total of the gathered data.
- On the other side of the coin, we have machine learning, which trains a system to perform complex tasks and uses harvested data and experience to become smarter.
- Data mining relies on vast stores of data (e.g., Big Data), which then, in turn, is used to make forecasts for businesses and other organizations.
- Machine learning, on the other hand, works with algorithms, not raw data.
- Data mining relies on human intervention and is ultimately created for use by people.
- Whereas machine learning's whole reason for existing is that it can teach itself and not depend on human influence or actions.
- Also, data mining is a process that incorporates two elements: **the database and machine learning**.
- The former provides data management techniques, while the latter supplies data analysis techniques.
- data mining can't learn or adapt, whereas that's the whole point with machine learning. Data mining follows pre-set rules and is static, while machine learning adjusts the algorithms as the right circumstances manifest themselves

MCQ

What is the primary goal of machine learning?
a) Minimize training time
b) Maximize model complexity
c) Minimize prediction accuracy
d) Maximize generalization to new data

Answer: d) Maximize generalization to new data

Which of the following is a type of unsupervised learning?
a) Regression
b) Classification
c) Clustering
d) Reinforcement learning

Answer: c) Clustering

What does the bias-variance tradeoff refer to in machine learning?
a) Balancing model complexity and interpretability
b) Balancing training time and prediction time
c) Balancing the model's ability to fit training data and generalize to new data
d) Balancing the tradeoff between precision and recall

Answer: c) Balancing the model's ability to fit training data and generalize to new data

MCQ

What is the main difference between supervised and unsupervised learning?
a) Supervised learning requires labeled data, while unsupervised learning does not.
b) Unsupervised learning requires labeled data, while supervised learning does not.
c) Supervised learning is used for classification, while unsupervised learning is used for regression.
d) Unsupervised learning is used for classification, while supervised learning is used for clustering.

Answer: a) Supervised learning requires labeled data, while unsupervised learning does not.

What is the purpose of a validation set in machine learning?
a) To train the model
b) To test the model
c) To fine-tune hyperparameters and avoid overfitting
d) To evaluate the model's performance on unseen data

Answer: c) To fine-tune hyperparameters and avoid overfitting

Which of the following is an example of a classification problem?
a) Predicting house prices
b) Identifying spam emails
c) Estimating stock prices
d) Forecasting temperature

Answer: b) Identifying spam emails